

Seita Kayukawa, Ph. D.

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Researcher at IBM Research

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Education

- Apr. 2020 - Sept. 2022** **Ph. D. of Engineering**
Graduate School of Advanced Science and Engineering, Waseda University
Advisor: Shigeo Morishima
- Apr. 2018 - Mar. 2020** **Master of Engineering**
Graduate School of Advanced Science and Engineering, Waseda University
Advisor: Shigeo Morishima
- Apr. 2014 - Mar. 2018** **Bachelor of Science**
Department of Applied Physics, Waseda University
Advisor: Shigeo Morishima

Work Experience

- Apr. 2023 - Current** **Researcher**
IBM Research
- Apr. 2021 - Mar. 2023** **Researcher**
Accessibility Lab., Miraikan - National Museum of Emerging Science and Innovation
- Apr. 2020 - Mar. 2023** **Research Fellow (~Sept. 2022: DC1, Oct. 2022~: PD)**
JSPS Research Fellowship for Young Scientists
- Feb. 2019 - Mar. 2020** **Research Intern**
IBM Research - Tokyo
- May 2018 - Sept. 2018** **Research Intern**
Cognitive Assistance Lab., Robotics Institute, Carnegie Mellon University

Research Interest

Human-Computer Interaction;

Real World Accessibility for People with Visual Impairments; Navigation; Non-visual Interface

Publications

Journal Papers and Conference Full Papers

- [1] Xiyue Wang, **Seita Kayukawa**, Hironobu Takagi, and Chieko Asakawa. 2025. **Engaging Visually Impaired People in Science Museums Through an Immersive Workshop: Practices, Challenges, and Opportunities**. In *Proceedings of the 27th International ACM SIGACCESS Conference on Computers and Accessibility (ASSETS '25)*.
DOI: <https://doi.org/10.1145/3663547.3746348>
- [2] Rie Kamikubo, **Seita Kayukawa**, Yuka Kaniwa, Allan Wang, Hernisa Kacorri, Hironobu Takagi, and Chieko Asakawa. 2025. **Beyond Omakase: Designing Shared Control for Navigation Robots with Blind People**. In *Proceedings of the 2025 CHI Conference on Human Factors in Computing Systems (CHI 2025)*.
DOI: <https://doi.org/10.1145/3706598.3714112>
- [3] Masaya Kubota*, Masaki Kuribayashi*, **Seita Kayukawa**, Hironobu Takagi, Chieko Asakawa, and Shigeo Morishima (* - equal contribution). 2024. **Snap&Nav: Smartphone-based Indoor Navigation System For Blind People via Floor Map Analysis and Intersection Detection**. In *Proceedings of the ACM on Human-Computer Interaction*, Volume 8, Issue MHCI, Article 275 (September 2024), 22 pages.
DOI: <https://doi.org/10.1145/3676522>
- [4] Yuka Kaniwa*, Masaki Kuribayashi*, **Seita Kayukawa**, Daisuke Sato, Hironobu Takagi, Chieko Asakawa, and Shigeo Morishima (* - equal contribution). 2024. **ChitChatGuide: Conversational Interaction Using Large Language Models for Assisting People with Visual Impairments to Explore a Shopping Mall**. In *Proceedings of the ACM on Human-Computer Interaction*, Volume 8, Issue MHCI, Article 247 (September 2024), 25 pages.
DOI: <https://doi.org/10.1145/3676492>
- [5] Xiyue Wang, **Seita Kayukawa**, Hironobu Takagi, Giorgia Masoero, and Chieko Asakawa. 2022. **Direct or Immersive? Comparing Smartphone-based Museum Guide Systems for Blind Visitors**. In *Proceedings of the 21st International Web for All Conference (W4A '24)*.
DOI: <https://doi.org/10.1145/3677846.3677856>
- [6] Xiyue Wang, **Seita Kayukawa**, Hironobu Takagi, and Chieko Asakawa. 2022. **TouchPilot: Designing a Guidance System that Assists Blind People in Learning Complex 3D Structures**. In *Proceedings of the 25th International ACM SIGACCESS Conference on Computers and Accessibility (ASSETS '23)*.
DOI: <https://doi.org/10.1145/3597638.3608426>
- [7] **Seita Kayukawa**, Daisuke Sato, Masayuki Murata, Tatsuya Ishihara, Hironobu Takagi, Shigeo Morishima, and Chieko Asakawa. 2023. **Enhancing Blind Visitor's Autonomy in a Science Museum Using an Autonomous Navigation Robot**. In *Proceedings of the 2023 CHI Conference on Human Factors in Computing Systems (CHI 2023)*.
DOI: <https://doi.org/10.1145/3544548.3581220>
- [8] Masaki Kuribayashi, Tatsuya Ishihara, Daisuke Sato, Jayakorn Vongkulbhisal, Karnik Ram, **Seita Kayukawa**, Hironobu Takagi, Shigeo Morishima, and Chieko Asakawa. 2023. **PathFinder: Designing a Map-less Navigation System for Blind People in Unfamiliar Buildings**. In *Proceedings of the 2023 CHI Conference on Human Factors in Computing Systems (CHI 2023)*.
DOI: <https://doi.org/10.1145/3544548.3580687>

- [9] Xiyue Wang, **Seita Kayukawa**, Hironobu Takagi, and Chieko Asakawa. 2022. **BentoMuseum: 3D and Layered Interactive Museum Map for Blind Visitors**. In *Proceedings of the 24th International ACM SIGACCESS Conference on Computers and Accessibility (ASSETS '22)*. DOI: <https://doi.org/10.1145/3517428.3544811>
- [10] **Seita Kayukawa**, Daisuke Sato, Masayuki Murata, Tatsuya Ishihara, Akihiro Kosugi, Hironobu Takagi, Shigeo Morishima, and Chieko Asakawa. 2022. **How Users, Facility Managers, and Bystanders Perceive and Accept a Navigation Robot for Visually Impaired People in Public Buildings**. In *Proceedings of the 31st IEEE International Conference on Robot & Human Interactive Communication (IEEE RO-MAN '22)*. DOI: <https://doi.org/10.1109/RO-MAN53752.2022.9900717>
- [11] Masaki Kuribayashi, **Seita Kayukawa**, Jayakorn Vongkulbhisal, Daisuke Sato, Chieko Asakawa, Hironobu Takagi, and Shigeo Morishima. 2022. **Corridor-Walker: Mobile Indoor Walking Assistance for Blind People to Avoid Obstacles and Recognize Intersections**. In *Proceedings of the ACM on Human-Computer Interaction*, Volume 6, Issue MHCI, Article 179 (September 2022), 22 pages. DOI: <https://dx.doi.org/10.1145/3546714>
- [12] Yutaro Yamanaka, **Seita Kayukawa**, Hironobu Takagi, Yuichi Nagaoka, Yoshimune Hiratsuka, and Satoshi Kurihara. 2021. **One-Shot Wayfinding Method for Blind People via OCR and Arrow Analysis with a 360-degree Smartphone Camera**. In *Proceedings of the 18th EAI International Conference on Mobile and Ubiquitous Systems: Computing, Networking and Services (MobiQuitous '21)*. DOI: https://doi.org/10.1007/978-3-030-94822-1_9
- [13] Masaki Kuribayashi*, **Seita Kayukawa***, Hironobu Takagi, Chieko Asakawa, and Shigeo Morishima (* - equal contribution). 2021. **LineChaser: A Smartphone-Based Navigation System for Blind People to Stand in Line**. In *Proceedings of the 2021 CHI Conference on Human Factors in Computing Systems (CHI '21)*. DOI: <https://doi.org/10.1145/3411764.3445451>
- [14] **Seita Kayukawa**, Tatsuya Ishihara, Hironobu Takagi, Shigeo Morishima, and Chieko Asakawa. 2020. **Guiding Blind Pedestrians in Public Spaces by Understanding Walking Behavior of Nearby Pedestrians**. In *Proceedings of the ACM on Interactive, Mobile, Wearable and Ubiquitous Technologies (IMWUT)*. Volume 4, Issue 3, Article 85 (September 2020), 22 pages. DOI: <https://doi.org/10.1145/3411825>
- [15] **Seita Kayukawa**, Keita Higuchi, João Guerreiro, Shigeo Morishima, Yoichi Sato, Kris Kitani, and Chieko Asakawa. 2019. **BBeep: A Sonic Collision Avoidance System for Blind Travellers and Nearby Pedestrians**. In *Proceedings of the 2019 CHI Conference on Human Factors in Computing Systems (CHI '19)*. DOI: <https://doi.org/10.1145/3290605.3300282>

Conference Short Papers, Demonstrations, and Posters

- [16] Hironobu Takagi, Kakuya Naito, Daisuke Sato, Masayuki Murata, **Seita Kayukawa**, and Chieko Asakawa. 2025. **Field Trials of Autonomous Navigation Robot for Visually Impaired People**. In *Proceedings of the Extended Abstracts of the 2025 CHI Conference on Human Factors in Computing Systems (CHI '25 Case Study)*.
DOI: <https://doi.org/10.1145/3706599.3706663>
- [17] **Seita Kayukawa**, Keita Higuchi, Shigeo Morishima, and Ken Sakurada. 2023. **3DMovieMap: An Interactive Route Viewer for Multi-Level Buildings**. In *Proceedings of the Extended Abstracts of the 2023 CHI Conference on Human Factors in Computing Systems (CHI '23 LBW)*.
DOI: <https://doi.org/10.1145/3544549.3585885>
- [18] Masaki Kuribayashi, **Seita Kayukawa**, Jayakorn Vongkulbhisal, Daisuke Sato, Chieko Asakawa, Hironobu Takagi, and Shigeo Morishima. 2021. **Designing a Smartphone-Based Assistance System for Blind People to Recognize Intersections and Obstacles in Indoor Corridors**. In *Proceedings of the 18th EAI International Conference on Mobile and Ubiquitous Systems: Computing, Networking and Services (MobiQuitous '21 Poster)*.
DOI: <https://doi.org/10.1007/978-3-030-94822-1>
- [19] **Seita Kayukawa**, Hironobu Takagi, João Guerreiro, Shigeo Morishima, and Chieko Asakawa. 2020. **Smartphone-Based Assistance for Blind People to Stand in Lines**. In *Proceedings of the Extended Abstracts of the 2020 CHI Conference on Human Factors in Computing Systems (CHI '20 LBW)*.
DOI: <https://doi.org/10.1145/3334480.3382954>
- [20] **Seita Kayukawa**, Tatsuya Ishihara, Hironobu Takagi, Shigeo Morishima, and Chieko Asakawa. 2020. **BlindPilot: A Robotic Local Navigation System that Leads Blind People to a Landmark Object**. In *Proceedings of the Extended Abstracts of the 2020 CHI Conference on Human Factors in Computing Systems (CHI '20 LBW)*.
DOI: <https://doi.org/10.1145/3170427.3189085>
- [21] Ryo Shimamura, **Seita Kayukawa**, Takayuki Nakatsuka, Shoki Miyagawa, and Shigeo Morishima. 2019. **A Study on the Sense of Burden and Body Ownership on Virtual Slope**. In *Proceedings of the IEEE Conference on Virtual Reality and 3D User Interfaces (IEEE VR '19 Poster)*.
DOI: <https://doi.org/10.1109/VR.2019.8797960>
- [22] **Seita Kayukawa**, Keita Higuchi, Ryo Yonetani, Masanori Nakamura, Yoichi Sato, and Shigeo Morishima. 2018. **Dynamic Object Scanning: Object-Based Elastic Timeline for Quickly Browsing First-Person Videos**. In *Extended Abstracts of the 2018 CHI Conference on Human Factors in Computing Systems (CHI '18 LBW and DEMO)*.
DOI: <https://doi.org/10.1145/3170427.3189085>

Awards

- Oct. 2024 Honorable Mention Award**
ASSETS 2025 (co authored paper, 1st author: Xiyue Wang)
- May 2024 Best Technical Paper Award**
W4A 2024 (co authored paper, 1st author: Xiyue Wang)
- Nov. 2021 Outstanding Student Paper Award**
MobiQuitous 2021 (co authored paper, 1st author: Yutaro Yamanaka)
- Dec. 2020 Best Paper Award**
JSSST WISS 2020 (a Domestic Conference in Japan)
- Mar. 2020 IPSJ Yamashita SIG Research Award**
Information Processing Society of Japan (IPSJ)
- Mar. 2019 Azusa Ono Memorial Award**
Waseda University
- Mar. 2019 Best Paper Award**
IPSJ Interaction 2019 (a Domestic Conference in Japan)

Scholarships

- May 2020 - Mar. 2021 Early Bird Program (Support for Young Researchers),
Waseda Research Institute for Science and Engineering**
- Apr. 2018 - Mar. 2020 JASSO Scholarship for Outstanding Master Students**
- May 2018 - Sept. 2018 Visiting Support from Super Global University**
- May 2018 - Aug. 2018 JASSO Scholarship for Short-term Study Abroad**

Skills

Programming Languages: C++, Python, Swift, HTML, CSS

Libraries / Platforms: OpenCV, Qt5, ROS, Arduino, CMake

OS: macOS, Ubuntu

Others: Adobe CC (Illustrator, Premiere Pro, Photoshop, InDesign)
User Studies, Statistical Analysis